

9. Two Stage Least Squares and Modern IV Estimators

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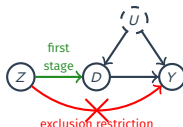
Jaewon (“Jay-one”) Yoo

National Tsing Hua University

Outline

Where are we? Where are we going?

- Last time:
 - **Instrumental variable** under noncompliance in randomized experiments.
 - The local ATE (or CACE): $\tau_{\text{LATE}} = \text{ITT}_{Y,\text{co}} = \frac{\text{ITT}_Y}{\text{ITT}_D}$
 - The Wald/IV estimator: $\widehat{\tau}_{\text{IV}} = \widehat{\text{ITT}}_Y / \widehat{\text{ITT}}_D$
 - Intent-to-treat analysis, compliance types, identification assumptions..



- Today:
 1. **Two-Stage Least Squares** (TSLS): estimands, weak IVs, multivalued D , general 2SLS.
 2. **Modern IV via Machine Learning**: PLIV (constant τ), DRIV (heterogeneous $\tau(\mathbf{X})$), DML.
 3. **Estimating IV in practice**: AJR walkthrough + modern IV in R.
 4. **Notable IV designs**: treatment assignments, regional characteristics, peers' environments, shift-share.



Source: *Chapter 4 of Mostly Harmless Econometrics (Textbook 1) by J. Angrist & J. Pischke*

1/ Two-Stage Least Squares

Two Stage Least Squares

- **Two-stage least squares** (TSLS) is the classical approach to IV which assumes two linear models:

$$Y_i = \alpha + \tau D_i + \varepsilon_i$$

$$D_i = \delta + \gamma Z_i + \eta_i$$

- Here, the treatment D_i is **endogenous** so $\mathbb{E}[\varepsilon_i | D_i] \neq 0$.
 - Canonical source: an omitted variable / unobserved confounder U opening a back-door path $D \leftarrow U \rightarrow Y$.
- But we have an **instrument** Z_i that is exogenous $\mathbb{E}[\varepsilon_i | Z_i] = 0$.
 - It's also exogenous for treatment uptake, so $\mathbb{E}[\eta_i | Z_i] = 0$.
- This implies the following CEF form for Y_i conditional on Z_i :

$$\mathbb{E}[Y_i | Z_i] = \alpha + \tau \mathbb{E}[D_i | Z_i] = \alpha + \tau \cdot (\gamma Z_i)$$

- α is reused loosely (technically $\alpha + \tau\delta$ here); not the parameter of our interest.

TSLS Estimands

- Under the model, we have the following CEF: $\mathbb{E}[Y_i|Z_i] = \alpha + \tau \cdot (\gamma Z_i)$
 - $\rightsquigarrow$ A regression of Y_i on γZ_i would have τ as the slope.

- If the CEF is linear, then we have this simple relationship slopes:

$$\mathbb{E}[D_i|Z_i] = \delta + \gamma Z_i \quad \rightsquigarrow \quad \gamma = \frac{\text{cov}(D_i, Z_i)}{\mathbb{V}(Z_i)}$$

- Applying this to above CEF, we have:

$$\tau = \frac{\text{cov}(Y_i, \gamma Z_i)}{\mathbb{V}(\gamma Z_i)} = \frac{\text{cov}(Y_i, Z_i)}{\gamma \mathbb{V}(Z_i)} = \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(D_i, Z_i)}$$

- TSLS estimator:
 - Estimate $\hat{\gamma}$ from regression of treatment D_i on instrument Z_i
 - Estimate $\hat{\tau}_{\text{TSLS}}$ as the slope of a regression of Y_i on $\hat{\gamma} Z_i$.
 - $\hat{D}_i = \hat{\delta} + \hat{\gamma} Z_i$; the Z -driven $\hat{\gamma} Z_i$ is exogenous, uncorrelated with ε_i (constant $\hat{\delta}$ doesn't affect the slope).
 - Under this model, $\hat{\tau}_{\text{TSLS}} \xrightarrow{P} \tau$ (but don't use SEs from second stage; see MHE section 4.6.1. 2SLS *Mistakes*)

Binary Treatment and Instrument

- Under binary treatment/instrument, TSLS estimand is the LATE:

$$\tau = \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(D_i, Z_i)} = \frac{\mathbb{E}[Y_i|Z_i = 1] - \mathbb{E}[Y_i|Z_i = 0]}{\mathbb{E}[D_i|Z_i = 1] - \mathbb{E}[D_i|Z_i = 0]} = \frac{\text{ITT}_Y}{\text{ITT}_D} = \tau_{\text{LATE}}$$

- $\text{cov}(Y, Z)/\text{cov}(D, Z) = \beta_Y/\beta_D$ ($\mathbb{V}(Z)$ cancels, regardless of Z). For binary Z , OLS slope $\beta_\cdot = \mathbb{E}[\cdot|Z=1] - \mathbb{E}[\cdot|Z=0]$ (W4), giving $\text{ITT}_Y/\text{ITT}_D$.
- LATE defined over compliers $\{D_i(1) > D_i(0)\}$, single group only for binary D .
- And that the TSLS estimator is the Wald estimator:

$$\widehat{\tau}_{\text{2SLS}} = \frac{\widehat{\text{cov}}(Y_i, Z_i)}{\widehat{\text{cov}}(D_i, Z_i)} = \frac{\overline{Y}_1 - \overline{Y}_0}{\overline{D}_1 - \overline{D}_0} = \frac{\widehat{\text{ITT}}_Y}{\widehat{\text{ITT}}_D} = \widehat{\tau}_{\text{iv}}$$

↪ Constant effects model is not required for TSLS in this setting.

TSLS with Covariates

- Adding covariates linearly:

$$Y_i = \alpha + \tau D_i + \mathbf{X}_i' \beta_y + \varepsilon_i$$

$$D_i = \delta + \gamma Z_i + \mathbf{X}_i' \beta_d + \eta_i$$

- This *uninteracted* form imposes constant τ across $\mathbf{X}$. Otherwise τ is an odd weighted function of causal effects and $\tau \neq \tau_{\text{LATE}}$.
- Why not Lin (W4) fully-interacted models ($\tau + \gamma' \tilde{\mathbf{X}}_i$ varies) in IV?
 1. Each $D_i \tilde{X}_{ij}$ is endogenous and requires its own instrument, e.g., $Z_i \tilde{X}_{ij}$.
 2. Strong prediction of D by Z does not imply strong prediction of $D_i \tilde{X}_{ij}$ by $Z_i \tilde{X}_{ij}$.
 3. Many interaction terms can amplify weak-IV problems and complicate interpretation.
- $\rightsquigarrow$ **PLIV** (§2) inherits constant τ (ML-flexible $g(\mathbf{X})$, $m(\mathbf{X})$ only). **DRIV** (§2) relaxes to $\tau(\mathbf{X})$.

Weak Instruments

- IV is unstable if instrument weakly affects treatment; $\text{cov}(D_i, Z_i) \approx 0$.
- **Example** completely irrelevant instrument:

$$\begin{aligned} Y_i &= \tau D_i + \varepsilon_i, & \mathbb{E}[\varepsilon_i | D_i] &\neq 0 \\ D_i &= 0 \times Z_i + \eta_i, & \mathbb{E}[\varepsilon_i | Z_i] = \mathbb{E}[\eta_i | Z_i] &= 0 \end{aligned}$$

- The bias of the Wald estimator:

$$\widehat{\tau}_{\text{IV}} - \tau = \frac{\widehat{\text{cov}}(\tau D_i + \varepsilon_i, Z_i)}{\widehat{\text{cov}}(D_i, Z_i)} - \tau = \frac{\frac{1}{n} \sum_{i=1}^n \varepsilon_i Z_i}{\frac{1}{n} \sum_{i=1}^n \eta_i Z_i} \xrightarrow{d} \underbrace{\frac{\text{cov}(\varepsilon_i, \eta_i)}{\mathbb{V}[\varepsilon_i]}}_{\text{bias}} + \underbrace{W_i}_{\text{Cauchy r.v.}}$$

- Inconsistent and asymptotically heavy tails (b/c of Cauchy).
 - When $Z \rightarrow D$ effect is small but non-zero, similar behavior.
- **Relevant IV**: noise divided by signal $\rightarrow 0$; **Irrelevant IV**: noise divided by noise does not.

What to Do About Weak Instruments?

- **Detect:** first-stage F-test on excluded instruments ($H_0 : \gamma = 0$).
 - $F \geq 10 \Rightarrow$ bias is small (Stock & Yogo 2005); still the dominant convention in empirical IV.
 - $F \geq 104.7$ for correct CI coverage in the worst-case DGP (Lee et al. 2022).
 - Lee et al. show the standard 1.96 gives true 95% coverage *only* at $F \geq 104.7$; below that, actual coverage drops monotonically. At the conventional $F = 10$, worst-case actual coverage is around **85%**.
- **Weak-IV robust inference:** Anderson-Rubin (1949) test (simplified setting, binary Z, D).
 - **Trick:** rewrite $H_0 : \tau = \tau_0$ as $H_0 : \text{ITT}_Y - \tau_0 \text{ITT}_D = 0$: **linearises the null**, no weak-IV pathology.
 - Under the null, asymptotically:

$$g(\tau_0) = \widehat{\text{ITT}}_Y - \tau_0 \widehat{\text{ITT}}_D \sim N(0, \Omega(\tau_0))$$
$$\Omega(\tau_0) = \mathbb{V}[\widehat{\text{ITT}}_Y] + \tau_0^2 \mathbb{V}[\widehat{\text{ITT}}_D] - 2\tau_0 \text{cov}(\widehat{\text{ITT}}_Y, \widehat{\text{ITT}}_D)$$

- AR statistic: $g(\tau_0)^2 / \Omega(\tau_0) \sim \chi_1^2$ regardless of first-stage strength; CI by analytical inversion.

```
1 > library(ivmodel)
2 > m <- ivmodel(Y = y, D = D, Z = Z)
3 > m$AR      # AR test (F-stat), p-value, and 95% CI by inversion
```

- Increasingly reported alongside the F-stat; handles weak-IV uncertainty correctly when first stage is borderline.

Multi-Valued Treatments

- Generalization of these ideas:
 - Multi-valued treatment: $D_i \in \{0, 1, \dots, K - 1\}$
 - Binary instrument: $Z_i \in \{0, 1\}$
- Assumptions:
 - Randomization: $[\{Y_i(z, d), \forall z, d\}, D_i(1), D_i(0)] \perp\!\!\!\perp Z_i$
 - Monotonicity: $D_i(1) \geq D_i(0)$ (instrument only increases treatment)
 - Exclusion restriction: $Y_i(1, d) = Y_i(0, d)$ for all $d = 0, 1, \dots, K - 1$
- Can't identify the proportion of all compliance types here.
- Example: $K = 3 \rightsquigarrow 9$ principal strata
 - Affected: $(D_i(0), D_i(1)) \in \{(0, 1), (0, 2), (1, 2)\}$
 - Unaffected: $(D_i(0), D_i(1)) \in \{(0, 0), (1, 1), (2, 2)\}$
 - Negatively affected: $(D_i(0), D_i(1)) \in \{(1, 0), (2, 0), (2, 1)\}$
 - Last ruled out by monotonicity.
 - 5 unknowns and 4 knowns under monotonicity (*see next slide*).

$K = 3$: Why Underidentified? Counting the Equations

Unknowns (under monotonicity): 6 strata $\rho_{00}, \rho_{01}, \rho_{02}, \rho_{11}, \rho_{12}, \rho_{22}$; sum to 1
 $\Rightarrow$ 5 free unknowns.

Knowns: observable $\mathbb{P}(D = d \mid Z = z)$ is a 2×3 table; each column sums to 1 $\Rightarrow$ 4 free moments.

What each moment “sees” (under monotonicity, off-diagonal strata vanish):

	$Z = 0$ (obs. $D_i(0)$)	$Z = 1$ (obs. $D_i(1)$)
$D = 0$	$\rho_{00} + \rho_{01} + \rho_{02}$	ρ_{00} ✓
$D = 1$	$\rho_{11} + \rho_{12}$	$\rho_{01} + \rho_{11}$
$D = 2$	ρ_{22} ✓	$\rho_{02} + \rho_{12} + \rho_{22}$

Two corner cells directly identify ρ_{00} and ρ_{22} . The remaining 4 unknowns ($\rho_{01}, \rho_{02}, \rho_{11}, \rho_{12}$) face 2 middle-row moments + 1 sum-to-1 = 3 equations
 $\Rightarrow$ 1-dim underidentified.

Punchline: at $Z = 1, D = 1$ we cannot tell *stayers-at-1* (ρ_{11}) from $0 \rightarrow 1$ compliers (ρ_{01}). One binary Z is not enough to separate them.

TSLS with Multivalued Treatments

- Let $C_i = jk$ be an indicator for compliance type $D_i(1) = j$ and $D_i(0) = k$.
 - People that are moved from k to j by the instrument.
 - Let $\rho_{jk} = \mathbb{P}(D_i(1) = j, D_i(0) = k)$ be the strata size.
- We can show that the 2SLS estimator converges to:

$$\hat{\tau}_{2SLS} \xrightarrow{p} \sum_{k=0}^{K-1} \sum_{j=k+1}^{K-1} \omega_{jk} \mathbb{E} \left(\frac{Y_i(1) - Y_i(0)}{j - k} \mid C_i = jk \right)$$
$$\omega_{jk} = \frac{(j - k)\rho_{jk}}{\sum_{s=0}^{K-1} \sum_{t=s+1}^{K-1} (s - t)\rho_{st}}$$

- Intuition: a weighted average of effects per dose for each affected type.
 - Weights are proportional to size of the strata and how big the effect of the instrument is for that strata.
 - If instrument can only increase by 1 dose, then simplifies to weighted average of principal strata effects.

2/ Modern IV via Machine Learning

Cf. Chernozhukov et al. (2018), *Double/Debiased Machine Learning*, Econometrics Journal

From Classical TSLS to Modern IV (lineage + ladder)

- Three modern IV-ML estimators relax classical TSLS-with- $\mathbf{X}$ (linear $\mathbf{X}$, constant τ):
- **IIVM** (Interactive IV Model; Chernozhukov et al. 2018, textbook Ch 13.2): ML extension of the *Wald estimator*. Binary Z , binary D . Targets the LATE on compliers via doubly-robust IPW score.
- **PLIV** (Partial Linear IV; Robinson 1988 original; Chernozhukov et al. 2018 adds DML): ML extension of TSLS-with- $\mathbf{X}$.

$$Y = \theta D + g(\mathbf{X}) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid \mathbf{X}, Z] = 0$$

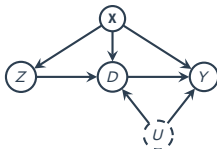
- $g(\cdot)$ ML-flexible; constant θ retained (D enters linearly). Applies to any D ; with binary D and heterogeneity, θ is a weighted average of conditional effects (close to but $\neq$ LATE).
- **DRIV** (Doubly Robust IV; Syrgkanis et al. 2019, *NeurIPS*): drop the constant- θ assumption $\rightsquigarrow$ heterogeneous $\theta(\mathbf{X})$.

$$Y = \theta(\mathbf{X}) D + g(\mathbf{X}) + \varepsilon$$

- D still linear-in- D ; coefficient $\theta(\mathbf{X})$ ML-learned. Doubly-robust orthogonal IV moment identifies $\theta(\cdot)$ pointwise.

ML Nuisance Functions and High-Dim Confounding

- Why ML for IV: $\mathbf{X}$ is high-dim (text, image embeddings, many interactions) and theory specifies neither features nor functional form.
- DAG with covariates $\mathbf{X}$ and unobserved confounder U :



- **Three nuisance functions** to estimate:

$$\ell_0(\mathbf{X}) = \mathbb{E}[Y \mid \mathbf{X}], \quad r_0(\mathbf{X}) = \mathbb{E}[D \mid \mathbf{X}], \quad m_0(\mathbf{X}) = \mathbb{E}[Z \mid \mathbf{X}]$$

- When $\mathbf{X} \not\rightarrow Z$ (i.e., $Z \perp \mathbf{X}$), m_0 collapses to the constant $\mathbb{E}[Z]$ and ML has nothing to do; it becomes a real function of $\mathbf{X}$ only when $\mathbf{X}$ predicts Z .
- **Why naive plug-in fails:** ML regularization injects bias into $\hat{\eta}$ that contaminates $\hat{\tau}$. DML fixes this with two ideas:
 - **Neyman orthogonality:** moment is locally insensitive to nuisance errors at the truth.
 - **Cross-fitting:** fit $\hat{\eta}$ on out-of-fold data, evaluate on held-out fold.

PLIV via DML: Algorithm and Conditions

- **Model:** $Y = \tau D + g(\mathbf{X}) + \varepsilon$, $\mathbb{E}[\varepsilon \mid \mathbf{X}, Z] = 0$.
- **Algorithm** (non-linear, high-dim Frisch-Waugh-Lovell): K -fold cross-fitting.
 1. Partition into folds $\{I_k\}_{k=1}^K$. For each k , fit $\hat{\ell}_{[k]}, \hat{r}_{[k]}, \hat{m}_{[k]}$ on data *outside* I_k .
 2. For $i \in I_k$: $\tilde{Y}_i = Y_i - \hat{\ell}_{[k]}(\mathbf{x}_i)$, $\tilde{D}_i = D_i - \hat{r}_{[k]}(\mathbf{x}_i)$, $\tilde{Z}_i = Z_i - \hat{m}_{[k]}(\mathbf{x}_i)$.
 3. Solve $\mathbb{E}_n[\tilde{Z}_i(\tilde{Y}_i - \tau \tilde{D}_i)] = 0$: **standard IV** on residuals.
- **Validity conditions** (CHS 2018, Thm 13.1.1): strong IV after partialling out, plus ML estimates converging at $n^{1/4}$ rate.
- Software: DoubleML (R). Applied PLIV demo on AJR in §3.

IIVM: LATE under One-Sided Noncompliance (book Ch 13.2)

- PLIV $Y = \theta D + g(\mathbf{X}) + \varepsilon$ applies to *any* D (binary or continuous); under heterogeneity its θ is a weighted average of conditional effects. With *binary* Z and D and noncompliance, the *exact LATE on compliers* is the textbook estimand of choice. **IIVM** (Interactive IV Model) targets the LATE directly via doubly-robust score with IPW reweighting.

- **Three nuisance regressions:**

$$\mu(z, \mathbf{X}) = \mathbb{E}[Y \mid Z = z, \mathbf{X}], \quad m(\mathbf{X}) = \mathbb{E}[D \mid \mathbf{X}], \quad p(\mathbf{X}) = \mathbb{P}(Z = 1 \mid \mathbf{X})$$

- Inverse-propensity weighting via $H(p) = \frac{Z}{p(\mathbf{X})} - \frac{1-Z}{1-p(\mathbf{X})}$ produces the doubly-robust LATE score.
- **Identification** uses the same assumptions as 2SLS LATE: random Z , exclusion, monotonicity, non-zero first stage. Under one-sided noncompliance the always-takers share is zero by construction (subgroups = list(always_takers = FALSE) in DoubleML).
- Software: **DoubleML::DoubleMLIIVM** (R/Py). Applied to BLI in §3 (lift

DML Weak-IV Robust Inference (book Ch 13.3–13.4)

- Anderson-Rubin (1949, §1) linearises the null and yields a χ^2 test that is valid for any first-stage strength. **DML generalises** this to high-dim $\mathbf{X}$ with cross-fitted ML nuisances.
- **Algorithm:** cross-fitted residuals $\check{Y}, \check{D}, \check{Z}$ from fold-out ML fits.
For each candidate θ on a grid:

$$M(\theta) = \mathbb{E}_n[(\check{Y} - \theta\check{D})\check{Z}], \quad V(\theta) = \mathbb{V}_n[(\check{Y} - \theta\check{D})\check{Z}], \quad C(\theta) = n \cdot$$

- $C(\theta) \stackrel{H_0}{\sim} \chi_1^2$ regardless of first-stage strength. 95% CR
= $\{\theta : C(\theta) \leq \chi_{0.95}^2(1) = 3.841\}$.
- No ratio, no Cauchy-tail pathology. The CR can be empty (rejection of *all* θ) or unbounded (very weak IV).
- **Two flavors:**
 - **PLIV-AR** (Ch 13.3): partial-linear, assumes constant θ .
 - **LATE-AR** (Ch 13.4): IIVM-style, targets the LATE on compliers under one-sided noncompliance.
- Returns to Fisher-style sharp-null testing (W_1 – W_2) on the IV null.
Applied to BLI in §3 (**star slide**).

From τ to $\tau(\mathbf{X})$: DRIV

- **DRIV** (Doubly Robust IV; Syrgkanis et al. 2019, *NeurIPS*): estimate the *function* $\tau(\mathbf{X})$ pointwise via doubly-robust orthogonal IV moment.
 - “How does the effect vary across covariates?”
 - “Doubly-robust” = the estimator stays consistent if either of two nuisance models is correctly specified.
 - **Implementation**: the NeurIPS authors released only a Python package (`EconML.DRIV`). R users call it from R via `reticulate`.

```
1 # DRIV in R via reticulate (loads EconML Python under the hood)
2 # One-time setup: reticulate::py_install("econml")
3 library(reticulate)
4 econml_dr <- import("econml.iv.dr")
5 sk_ens    <- import("sklearn.ensemble")
6
7 est <- econml_dr$DRIV(
8   model_y_xw = sk_ens$RandomForestRegressor(), # E[Y | X]   nuisance for outcome
9   model_t_xw = sk_ens$RandomForestClassifier(), # E[D | X]   nuisance for treatment
10  model_z_xw = sk_ens$RandomForestClassifier(), # E[Z | X]   nuisance for instrument
11  cv = 5L                                     # K-fold cross-fitting
12 )
13 est$fit(Y = D, Z = Z, X = X)                  # cross-fitted nuisances + orthogonal moment
14 est$effect(X_new)                             # tau-hat(X): heterogeneous effect at any X
```

- **Validating heterogeneity**: does $\tau(\mathbf{X})$ actually vary? **Best Linear Projection** (terminology and inferential framework: Semenova–Chernozhukov 2021, *Econometrics J.*) regresses $\hat{\tau}(\mathbf{X})$ on $\mathbf{X}$ with valid SE; the GRF function `best_linear_projection` implements it. Calibration tests check whether $\hat{\tau}(\mathbf{X})$ varies above sampling noise. Applied in §3.

DRIV vs. Causal Forest IV: Distinct R-side and Python-side Estimators

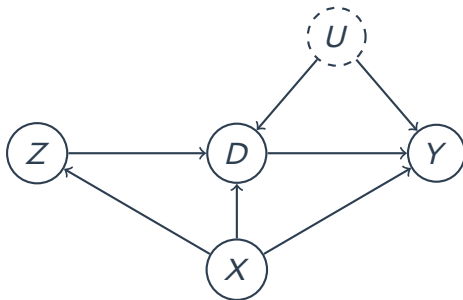
- **Implementations:** DRIV proper is Python-only (`econml.DRIV`, `econml.LinearDRIV`, `econml.ForestDRIV`). The R-side workhorse for HTE-IV is a *distinct* estimator, `grf::instrumental_forest` (Causal Forest IV).
- Same target heterogeneous treatment effect (DRIV literature writes $\theta(\mathbf{X})$, GRF literature writes $\tau(\mathbf{X})$), different moment construction:
 - DRIV uses the doubly-robust *orthogonal* moment with cross-fitted nuisances $(\hat{g}, \hat{r}, \hat{p})$.
 - Causal Forest IV uses honest forest weights on a *local* 2SLS moment (Athey–Tibshirani–Wager 2019).
- **Coming in §3:** classical AJR walkthrough, then applied HTE-IV demo on a digital advertising experiment ($n \approx 1.4\text{M}$, one-sided noncompliance). §3 runs *both* DRIV (`econml` via `reticulate`) and Causal Forest IV (R `grf`); they agree on the substantive HTE drivers.

3/ Estimating IV: AJR Walkthrough + Modern IV

From Theory to Practice: Classical TSLS Setup

- Modern IV (§2) shines when $\mathbf{X}$ is high-dim or text-scale. AJR sits in the **classical corner**: a single binary IV, modest $\mathbf{X}$; classical TSLS suffices.
- Classical-corner ingredients:
 1. observational data with *unobserved* confounders U ;
 2. a single instrumental variable Z with a qualitatively defensible exclusion restriction;
 3. low-dim observed covariates $\mathbf{X}$; (linearly conditioned).
- DAG (recap from §2): see next slide.

Classical TSLS Setup: DAG (recap from §2)



- $Z \rightarrow D$: first stage. $D \rightarrow Y$: causal effect of interest. U : unobserved confounder. X : observed covariates.
- Exclusion restriction: Z affects Y only through D (no direct $Z \rightarrow Y$ arrow).

Property Rights & Economic Development



Created using DALL-E 3 motivated by Ratna Sagar Shrestha



Recognize the person on the right?

- Q: Do property rights (i.e., institutions) promote economic development?
 - Famous paper on this: **Acemoglu, Johnson, and Robinson (2001) AER**
 - Relationship between strength of property rights in a country and GDP.

The AJR Data

Name	Description
shortnam	three-letter country code
africa	indicator for if the country is in Africa
asia	indicator for if country is in Asia
logem4	log mortality rates faced by European settlers (IV)
avexpr	strength of property rights (protection against expropriation)
logpgp95	log GDP per capita

```
1 > ajr <- read_csv("https://bit.ly/3RUJDWK"); ajr
2
3 # A tibble: 163 × 15
4   shortnam africa lat_abst malfal94 avexpr logpgp95 logem4 asia yellow baseco leb95
5   <chr>      <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
6 1 AFG          0 0.367 0.00372 NA      NA      4.54 1 0 NA NA
7 2 AGO          1 0.137 0.950 5.36 7.77 5.63 0 1 1 46.5
8 3 ARE          0 0.267 0.0123 7.18 9.80 NA 1 0 NA NA
9 4 ARG          0 0.378 0 6.39 9.13 4.23 0 0 1 72.9
10 5 ARM          0 0.444 0 NA 7.68 NA 1 0 NA NA
11 6 AUS          0 0.300 0 9.32 9.90 2.15 0 1 1 78.2
12 7 AUT          0 0.524 0 9.73 9.97 NA 0 0 NA NA
13 8 AZE          0 0.448 0 NA 7.31 NA 1 0 NA NA
14 9 BDI          1 0.0367 0.950 NA 6.57 5.63 0 1 NA NA
15 10 BEL          0 0.561 0 9.68 9.99 NA 0 0 NA NA
16 # i 153 more rows
17 # i 4 more variables: imr95 <dbl>, meantemp <dbl>, lt100km <dbl>, latabs <dbl>
18 # i Use `print(n = ...)` to see more rows
```

In R: Example Code Using ajr Data

```
1 # Center (i.e., demeaning) the variables
2 > ajr <- ajr |>
3   mutate(
4     D_cnt = avexpr - mean(avexpr, na.rm = TRUE),
5     Y_cnt = logpgp95 - mean(logpgp95, na.rm = TRUE),
6     Z_cnt = logem4 - mean(logem4, na.rm = TRUE)
7   ) |>
8   na.omit()
9
10 # Compute the ITTs on D and on Y:
11 > ITT_D <- cov(ajr$Z_cnt, ajr$D_cnt)
12 > ITT_Y <- cov(ajr$Z_cnt, ajr$Y_cnt)
13 > wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
14 [1] 0.9242
15
16 # Same as reg Y~Z / reg D~Z:
17 > ITT_Y <- coef(lm(Y_cnt ~ Z_cnt, data = ajr))[[2]]
18 > ITT_D <- coef(lm(D_cnt ~ Z_cnt, data = ajr))[[2]]
19 > wald_estimate <- ITT_Y / ITT_D; round(wald_estimate, digits = 4)
20 [1] 0.9242
```

- The Wald estimate from manual calculation: 0.9242

In R: Example Code Using ajr Data

```
1 # Compare with ivreg
2 > ivreg_result <- AER::ivreg(Y_cnt ~ D_cnt | Z_cnt, data = ajr); summary(ivreg_result)
3
4 Call:
5 AER::ivreg(formula = Y_cnt ~ D_cnt | Z_cnt, data = ajr)
6
7 Residuals:
8      Min       1Q   Median       3Q      Max
9 -2.40175 -0.54950  0.01792  0.68944  1.67361
10
11 Coefficients:
12             Estimate Std. Error t value Pr(>|t|)
13 (Intercept) 3.031e-16  1.231e-01   0.000      1
14 D_cnt       9.242e-01  1.547e-01   5.974 1.59e-07 ***
15 ---
16 Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
17
18 Residual standard error: 0.9458 on 57 degrees of freedom
19 Multiple R-Squared: 0.1107,      Adjusted R-squared: 0.09506
20 Wald test: 35.68 on 1 and 57 DF, p-value: 1.589e-07
21
22 > round(coef(ivreg_result), digits = 4)
23 (Intercept)      D_cnt
24      0.0000      0.9242
```

- The 2sls estimate from AER::ivreg(): 0.9242
 - Caveat: **compute robust SE separately!** (don't use SE from 2nd stage)

In R: Example Code Using ajr Data

```
1 # Estimate robust SE with estimatr::iv_robust()
2 > iv_rob <- estimatr::iv_robust(Y_cnt ~ D_cnt | Z_cnt, data = ajr, se_type = "HC2"); iv_rob
3           Estimate Std. Error      t value      Pr(>|t|)      CI Lower CI Upper DF
4 (Intercept) 4.148820e-16  0.1234908 3.359619e-15 1.000000e+00 -0.2472860 0.247286 57
5 D_cnt       9.242173e-01  0.1791511 5.158871e+00 3.262823e-06  0.5654735 1.282961 57
```

- Robust SE with HC2 using `estimatr::iv_robust()`: 0.1791

```
1 > iv_rob_cov <- estimatr::iv_robust(Y_cnt ~ D_cnt + lat_abst + asia + africa |
2                                   Z_cnt + lat_abst + asia + africa, data = ajr,
3                                   se_type = "HC2")
4 > var_labels <- c(
5   "D_cnt" = "Avg. Expropriation Risk (D_cnt)", "lat_abst" = "Abs. Value of Latitude",
6   "asia" = "Asian country", "africa" = "African country", "Z_cnt" = "log Mortality Rate (D_cnt)"
7 )
```

- When adjusting for covariates in TSLS, include them both in the 1st and 2nd stage.

IV Regression Table with model summary

```
1 > modelsummary::modelsummary(  
2   models = list(ivreg_result, iv_robust, iv_rob_cov),  
3   coef_map = var_labels, gof_map = c("nobs", "r.squared", "adj.r.squared"), stars = T,  
4   notes = "Note: See appendix for other model statistics.", output = "modelsummary_tab.tex")
```

	(1)	(2)	(3)
Avg. Expropriation Risk (D_cnt)	0.924*** (0.155)	0.924*** (0.179)	1.015** (0.351)
Abs. Value of Latitude			-1.596 (1.529)
Asian country			-1.048* (0.425)
African country			-0.390 (0.342)
Num.Obs.	59	59	59
R2	0.111	0.111	0.045
R2 Adj.	0.095	0.095	-0.026

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: See appendix for other model statistics.

Modern IV in R: DoubleML for PLIV

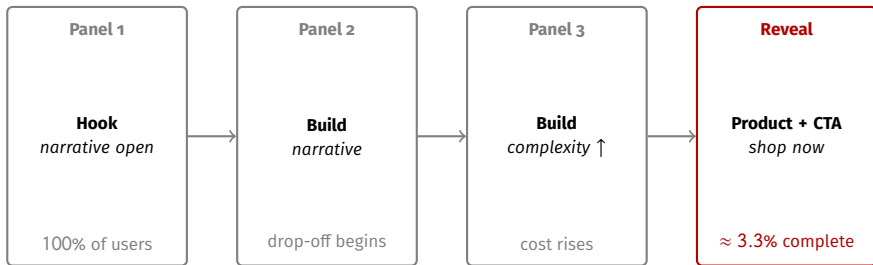
- Same partialling-out as Modern IV section, in ~ 10 R lines (cross-fitting built in):

```
1 > library(DoubleML); library(mlr3); library(mlr3learners)
2 > # Build DoubleMLData (y = outcome, d = treatment, z = instrument, x = covariates)
3 > data <- DoubleMLData$new(df, y_col = "Y", d_cols = "D",
4 +                           z_cols = "Z", x_cols = X_names)
5 > # ML learners for the nuisance functions
6 > ml_l <- lrn("regr.ranger") # E[Y | X]
7 > ml_m <- lrn("regr.ranger") # E[Z | X]
8 > ml_r <- lrn("regr.ranger") # E[D | X]
9 > # PLIV with 5-fold cross-fitting (Neyman orthogonality + cross-fitting = DML)
10 > pliv <- DoubleMLPLIV$new(data, ml_l, ml_m, ml_r, n_folds = 5)
11 > pliv$fit(); pliv$summary()
```

- Heterogeneous $\tau(\mathbf{X})$ in R: `grf::instrumental_forest` (Athey–Tibshirani–Wager 2019).
- **Bad controls:** DML does *not* pick confounders for you. ML selects by predictive power; colliders/mediators in $\mathbf{X} \rightsquigarrow$ biased $\hat{\tau}$ (Hünermund, Louw & Caspi 2023, *JCI*). **DAG reasoning (ECI W6) still your job.**
- Also: $\hat{\tau}$ sensitive to learner / $\mathbf{X}$ choice (Ahrens et al. 2025 §5–6, by DML 2018 authors). Remedies proposed; revisit in **W15**.

What Does a BLI Ad Look Like?

- Blind Lead-In (BLI) ads are multi-stage interactive sliders. The product is revealed *only* on the last panel; users must advance through each panel themselves.



- In passive media (TV, pre-roll), exposure $\Rightarrow$ reveal automatically. In BLI, only $\approx 3.3\%$ of exposed users reach the reveal.
- $\Rightarrow$ The behavioral object of interest is D (completion), not Z (exposure). Standard ITT-on- Z averages over 96.7% never-takers.

Modern IV Demo: A Digital Advertising Experiment

- Online book retailer A/B-tests two ad formats ($n \approx 1.4\text{M}$ user-impressions, 76 ad creatives).
 - Z : random assignment to interactive (Blind Lead-Ins, BLI) vs. static banner.
 - D : BLI *completion* (treated-side completion rate $\approx 3.32\%$; **one-sided noncompliance**: $D = 0$ for all $Z = 0$).
 - Y : 30-day purchase conversion (binary).
- Why this is a useful DML-IV testbed:
 - High-dim creative features (face presence, slide count, character count, color, area ratios).
 - Rich user covariates (spending, visits, age, gender, B2B, genre fit).
 - Plausible $\tau(\mathbf{X})$ heterogeneity along creative-content $\times$ user-engagement axes.
- **Plan**: (i) classical pooled 2SLS LATE (fixest); (ii) Anderson-Rubin robust CR (DML weak-IV inference, book Ch 13.3–13.4); (iii) DRIV via reticulate (econml.LinearIntentToTreatDRIV) for $\hat{\theta}(\mathbf{X})$; (iv) Causal Forest IV (grf::instrumental_forest) for nonparametric $\hat{\tau}(\mathbf{X})$. *Two distinct HTE-IV estimators on the same data; do they agree?*

Pooled 2SLS LATE: Classical Baseline

- 2SLS LATE on 30-day conversion (date FE, user covariates, cluster-robust SE at the user level):

```
1 > library(fixest)
2 > late <- feols(lngt_conv ~ ctrl_vars | date | tot ~ treatment,
3 +             data = d, cluster = ~ memno)
4 > summary(late)
5 TLS estimation, Dep. Var.: lngt_conv
6 Observations: 1,408,121 / Fixed-effects: date (80)
7 Standard-errors: Clustered (memno)
8             Estimate Std. Error t value Pr(>|t|)
9 fit_tot 0.002326    0.001210   1.922   0.0548 .
10 ... (control coefficients omitted)
11 First-stage F (Kleibergen-Paap): ~1.5e4 # very strong instrument
```

- LATE:** BLI completion raises 30-day conversion by ≈ 0.23 **percentage points** among compliers ($\text{fit_tot} = 0.00233$, $p \approx 0.055$).
- Weak-IV diagnostic:** first-stage $F = 10,007$ (Kleibergen-Paap, cluster-robust). Compliance is small (3.32%) but big-tech ad RCTs are typically large- N enough that the first stage is overwhelming; Stock-Yogo $F \geq 10$ **vastly** cleared. *AR robust CR is becoming standard reporting practice — check next slide.*

Anderson-Rubin Robust CR on the BLI Data

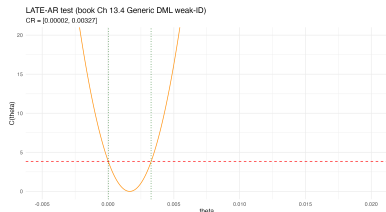
- Same test as §1 (linearise null) and §2 (DML grid inversion), applied to BLI.

- Cross-fitted DML residuals $\check{Y}, \check{D}, \check{Z}$. For each θ : $C(\theta) = n \cdot \bar{M}(\theta)^2 / \bar{V}(\theta)$, $\bar{M}(\theta) = \mathbb{E}_n[(\check{Y} - \theta \check{D}) \check{Z}]$.
- 95% CR
= $\{\theta : C(\theta) \leq \chi^2_{0.95}(1) = 3.841\}$. No denominator $\Rightarrow$ no Cauchy / heavy-tail pathology.

- BLI result (LATE-AR, book Ch 13.4):

Method	95% CR	Excludes o?
PLIV-AR (Ch 13.3)	$[-0.0029, 0.0040]$	No
LATE-AR (Ch 13.4)	$[+0.00002, 0.00327]$	Yes
Standard 2SLS Wald CI	$[+0.00007, 0.00451]$	Yes

- LATE-AR CR almost matches standard CI here (pooled $F = 10,007$). AR's value dominates when first stage is borderline.



LATE-AR $C(\theta)$ vs. $\chi^2_{0.95}(1) = 3.841$ cutoff. CR = θ values below cutoff.

DRIV via Reticulate (Syrngkanis et al. 2019)

- DRIV: heterogeneous LATE $\theta(X) = \alpha + \beta'X$ via cross-fitted nuisance + doubly-robust orthogonal score. Implementation:
`econml.LinearIntentToTreatDRIV` (Python only). Call from R via `reticulate`:

```
1 library(reticulate)
2 econml_iv <- import("econml.iv.dr"); sk <- import("sklearn.ensemble")
3 est <- econml_iv$LinearIntentToTreatDRIV(
4   model_y_xw = sk$RandomForestRegressor(n_estimators = 100L),
5   model_t_xwz = sk$RandomForestClassifier(n_estimators = 100L),
6   flexible_model_effect = "auto", cv = 5L, cov_clip = 0.005)
7 est$fit(Y = Y, T = D, Z = Z, X = X_paper, W = W_paper)
8 est$ate__inference()$summary_frame()
```

- ATE across `cov_clip` sensitivity (key trimming knob for low-compliance cells):

cov_clip	ATE	s.e.	95% CI	Note
0.001	0.00197	0.00122	[-0.0004, +0.0044]	most permissive
0.005	0.00247	0.00107	[+0.0004, +0.0046]	book default
0.010	0.00209	0.00078	[+0.0006, +0.0036]	
0.050	0.00056	0.00019	[+0.0002, +0.0009]	over-trimmed

- Linear projection $\beta'X$ flags which features drive $\hat{\theta}(X)$ heterogeneity. With face-stage decomposition (`first_has_face`, `middle_has_face`, `late_has_face`): `middle_has_face` is the only significant face term $\Rightarrow$ middle-panel face matters causally. Side-by-side with Causal Forest IV in the next slide.

From $\hat{\tau}$ to $\hat{\tau}(\mathbf{X})$: Causal Forest IV in R

- R-native HTE-IV (distinct from DRIV; §2): `grf::instrumental_forest` (Athey–Tibshirani–Wager 2019); $\mathbf{X}$ = 4 user + 4 creative features.

```
1 > library(grf)
2 > ivf <- instrumental_forest(
3 +   X = X_matrix, Y = Y, W = D, Z = Z,
4 +   num.trees = 4000, min.node.size = 100, honesty = TRUE)
5 > average_treatment_effect(ivf)      # forest-weighted ATE on BLI sample
6 > predict(ivf, X_new)$predictions    # tau-hat(X) at any covariate point
7 > best_linear_projection(ivf, A = X_matrix) # which features drive HTE; valid SE
8 > test_calibration(ivf)               # is HTE detectable above noise?
```

- Causal Forest ATE:** ≈ 0.17 pp on average (forest-weighted; conservative vs. pooled LATE because low-compliance cells are down-weighted).
- BLP** marginal HTE drivers: `face_early` ($p \approx 0.07$), `pref_matched` ($p \approx 0.10$) $\Rightarrow$ creative-content $\times$ user-engagement axes.
- Same target $\hat{\tau}(\mathbf{X})$ as DRIV (§2), built from forest-weighted local 2SLS rather than the orthogonal-score moment.

What HTE-IV Reveals on the BLI Data

- **Pooled-LATE ladder converges:** 2SLS 0.00229, DML PLIV (Lasso) 0.00217, DML IIVM (RF) 0.00164 ($p = 0.049$), DRIV (econml, cov_clip=0.005) 0.00247. All in the $[+322\%, +484\%]$ lift band $\Rightarrow$ the BLI-completion effect on conversion is robust to ML, score, and functional-form choice.
- **DRIV and Causal Forest IV both flag middle-panel face presence:**
 - middle_has_face (positive): Causal Forest BLP $\hat{\beta} = +0.0144$ ($p = 0.008$); DRIV linear $\hat{\beta} = +0.0118$ ($p = 0.014$). *Both significant.*
 - late_has_face (negative): Causal Forest $\hat{\beta} = -0.0103$ ($p = 0.086$, marginal); DRIV $\hat{\beta} = -0.005$ ($p = 0.33$, not significant). The forest picks up the late-stage penalty more sharply.
 - Engaged users (high recent visit-days) absorb ≈ 0.39 pp LATE; disengaged ≈ 0.05 pp $\Rightarrow$ near-8 $\times$ heterogeneity gap pooled 2SLS hides.
- **Mapping back to the paper:** H4 (face presence raises completion) confirmed at the *conversion-LATE* level. Stage asymmetry (middle dominant, late negative) is now *causally identified* on the downstream outcome, not just a descriptive completion-stage association.

Lift Translation: from $\hat{\tau}$ to Industry Magnitude

- Gordon, Zettelmeyer, Bhargava & Chapsky 2019, *Marketing Science* eq. (8): lift = incremental conversion among treated, expressed as a percentage of the treated group's *counterfactual* (untreated) conversion.

$$\tau_{\ell} = \frac{\Delta \text{Conversion}_{\text{treated}}}{\mathbb{E}[Y^{obs} \mid Z = 1, W^{obs} = 1] - \tau} = \frac{\tau}{\mathbb{E}[Y^{obs} \mid Z = 1, W^{obs} = 1] - \tau}$$

- Under random assignment, the denominator is identified by the control-group conversion. NNE = $1/\text{ITT}_Y = 1/(\text{compliance rate} \times \hat{\tau}_{\text{LATE}})$: ad impressions per incremental purchase.
- BLI lift table (control conversion 0.051% as estimator of treated-counterfactual baseline):

Method	$\hat{\theta}$	Compliers conv.	Lift (%)	NNE
2SLS pooled (paper)	0.00229	0.280%	+449%	13,200
DML PLIV (Lasso)	0.00217	0.268%	+426%	13,900
DML IIVM (RF)	0.00164	0.215%	+322%	18,400
DRIV (econml, cov_clip 0.005)	0.00247	0.298%	+484%	12,300

- Industry benchmark: typical online display-ad NNE is 50,000–200,000. BLI's 12,300–18,400 is order-of-magnitude better.
- **Back-of-envelope:** 100M annual impressions $\times$ 3.32% completion $\times \hat{\tau}_{\text{LATE}} (\approx 0.0023) \approx 7,640$ **extra purchases/year** causally attributable to BLI completion ($\approx 114\text{M}$ KRW at the retailer's average book price).

4/ Notable IV Designs in Recent Empirical Work

When you have a great idea for an IV but the first stage turns out to be not significant.

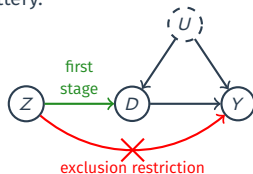


Source: [Causal Inference for the Brave and True](#) By Matheus Facure Alves

Treatment Assignments as IVs

Bloom et al. (QJE 2015)¹

- Context: 9-month RCT in Ctrip's Shanghai call-centre; 249 volunteers entered the lottery.



Instrument Z: even-birthday lottery eligibility.

Treatment D: actually working from home ≥ 4 days/week.

Outcome Y: weekly productivity index (calls, minutes logged-in, calls/minute).

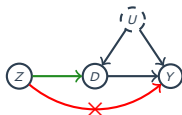
- Identification issue: Endogenous uptake; one-sided non-compliance (some winners revert, losers cannot WFH).
- IV logic:
 1. Randomization: Balance tests on 18 baseline characteristics show no joint differences; $p=0.466$.
 2. Relevance: 80-90% of winners comply (1st-stage $F \approx 23$)
 3. Exclusion: tasks, pay, IT identical – only location changes.
 4. Monotonicity: winners may or may not take WFH, losers cannot $\rightsquigarrow$
 $D_i(1) \geq D_i(0)$ trivially (one-sided non-compliance).

¹ Bloom, N., Liang, J., Roberts, J. & Ying, Z. "Does Working from Home Work?" QJE 2015. 130(1): 165–218.

Regional Characteristics as IVs

Narang and Shankar (*Marketing Sci.* 2019)²

- Context: Observational panel on 32 mil. customers of a U.S. video-game / electronics retailer, Jan 2013 – Dec 2015; firm's mobile-shopping app launched July 2014.



Instrument Z: number of cell towers in shopper's ZIP.

Treatment D: adoption of the retailer's mobile app.

Outcome Y: Monthly on/offline purchase & return amounts.

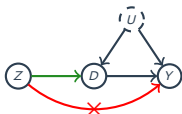
- Identification issue: Self-selection. Tech-savvy or high-value shoppers more likely to install the app.
- IV logic:
 1. Randomization: 90% of cell-towers built before 2010 and shows only weak correlations with 2010–15 population growth and store count, included in $\mathbf{X}_i$.
 2. Relevance: Each cell tower raises the log-odds of adoption by 0.0022 ($z=3.7$).
 3. Exclusion: controlled for ZIP income, store distance, and competitor presence to block direct demand channels.
 4. Monotonicity: Better signal only increases the prob. of downloading the app (no defiers).

² Narang & Shankar. "Mobile App Introduction and On- /Off-line Purchases and Product Returns." *Marketing Sci.* 2019. 38(5): 756 – 772.

Peers' Environments as IVs

Aral and Nicolaides (*Nature Comm.* 2017)³

- Context: Global fitness-tracking network, 1.1 Mil. runners, 3.4 Mil. ties, 350 Mil. km logged over 5 years.



Instrument Z: daily weather (rain/extreme temperature) in friend's city.

Treatment D: friend's running distance that day.

Outcome Y: user/ego's running distance (same or next day).

- Identification issue: Homophily & shared shocks. Similar friends exercise together or face the same local weather.
- IV logic:
 1. Randomization: Use only friend pairs in different cities whose weather paths are uncorrelated; ego's own weather + date FE included.
 2. Relevance: first-stage $F = 216-430$ well above Stock-Yogo cutoff.
 3. Exclusion: Weather in friend's city is uncorrelated with ego's weather by construction.
 4. Monotonicity: bad weather never makes the friend run more.

³ Sinan A. & Nicolaides C. "Exercise contagion in a global social network." *Nature Communications*. 2017. 8: 14753.

Onto the presentations & discussions!

Contact Information:

jaewon.yoo@iss.nthu.edu.tw

<https://j1yoo.github.io/>



Appendix

BLI Demo: Method-Comparison Table

Method	$\hat{\theta}$	s.e.	95% CI	Notes
2SLS pooled (paper)	0.00229	0.00121	[-0.00004, 0.00469]	fixest, date FE, cluster=memno
DML PLIV (Lasso)	0.00217	0.00116	[-0.00010, 0.00444]	Book Ch 13.1 score, $n_{\text{rep}} = 5$
DML PLIV (RF, m=featureless)	0.00137	0.00190	[-0.00235, 0.00509]	Book Ch 13.1
DML IIVM (Lasso)	0.00217	0.00116	[-0.00009, 0.00444]	Book Ch 13.2
DML IIVM (RF)	0.00164	0.00083	[+0.00001, 0.00328]	p = 0.049
DRIV linear [cov_clip 0.001]	0.00197	0.00122	[-0.00041, 0.00435]	econml.LinearIntentToTreatDRIV
DRIV linear [cov_clip 0.005]	0.00247	0.00107	[+0.00037, 0.00458]	book default
DRIV linear [cov_clip 0.010]	0.00209	0.00078	[+0.00057, 0.00362]	
AR robust CR (LATE-AR Ch 13.4)	—	—	[+0.00002, 0.00327]	<i>robust to weak IV</i>

- All point estimates in [0.00164, 0.00247], lift band [+322%, +484%] over 0.051% control conversion.
- Working paper identification + inference still under revision; values shown are illustrative for the lecture demo.

BLI Demo: Subgroup CATE (engaged vs disengaged users)

Subgroup	<i>n</i>	CATE
Engaged (high recent visit-days)	686,903	0.00222
Disengaged (low recent visit-days)	721,218	0.00114
Age ≥ 30	932,236	0.00146
Age < 30	475,885	0.00207
face_early present	586,313	0.00235
face_early absent	821,808	0.00118

- Engaged $\times$ face-early users absorb *nearly all* the BLI completion lift; disengaged users with no early face: near-zero CATE.
- Source: `grf::instrumental_forest` subgroup CATE; trimmed cells with compliance $< 0.5\%$.

General 2SLS

- Linear model for each i :

$$Y_i = \mathbf{X}_i' \boldsymbol{\beta} + \varepsilon_i$$

- $\mathbf{X}_i$ is $k \times 1$ and now includes D_i and any pretreatment covariates.
- Parts of $\mathbf{X}_i$ are endogenous so that $\mathbb{E}[\varepsilon_i | \mathbf{X}_i] \neq 0$
- Instruments $\mathbf{Z}_i$ that is $\ell \times 1$ vector such that $\mathbb{E}[\varepsilon_i | \mathbf{Z}_i] = 0$.
 - $\mathbf{Z}_i$ might include exogenous/pretreatment variables from $\mathbf{X}_i$ as well.
 - Rank condition: $\mathbb{E}[\mathbf{Z}_i \mathbf{Z}_i']$ and $\mathbb{E}[\mathbf{X}_i \mathbf{Z}_i']$ have full rank.
- Identification:
 - $k = \ell$: just-identified.
 - $k < \ell$: over-identified (can test the exclusion restriction, kinda)
 - $k > \ell$: unidentified (fails rank condition)

Nasty Matrix Algebra

- Projection matrix projects values of $\mathbf{X}_i$ onto $\mathbf{Z}_i$:

$$\mathbf{\Pi} = (\mathbb{E}[\mathbf{Z}_i \mathbf{Z}_i'])^{-1} \mathbb{E}[\mathbf{Z}_i \mathbf{X}_i'] \quad (\text{projection matrix, i.e., } \mathbf{\Pi} / \mathbf{\Pi})$$

$$\tilde{\mathbf{X}}_i = \mathbf{\Pi}' \mathbf{Z}_i \quad (\text{projected values})$$

- To derive the 2SLS estimator, take the fitted values, $\mathbf{\Pi}' \mathbf{Z}_i$ and multiply both sides of the outcome equation by them:

$$Y_i = \mathbf{X}_i' \boldsymbol{\beta} + \varepsilon_i$$

$$\mathbf{\Pi}' \mathbf{Z}_i Y_i = \mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i' \boldsymbol{\beta} + \mathbf{\Pi}' \mathbf{Z}_i \varepsilon_i$$

$$\mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i Y_i] = \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i'] \boldsymbol{\beta} + \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \varepsilon_i]$$

$$\mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i Y_i] = \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i'] \boldsymbol{\beta} + \mathbf{\Pi}' \mathbb{E}[\mathbf{Z}_i \varepsilon_i]$$

$$\mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i Y_i] = \mathbb{E}[\mathbf{\Pi}' \mathbf{Z}_i \mathbf{X}_i'] \boldsymbol{\beta}$$

$$\mathbb{E}[\tilde{\mathbf{X}}_i Y_i] = \mathbb{E}[\tilde{\mathbf{X}}_i \mathbf{X}_i'] \boldsymbol{\beta}$$

$$\boldsymbol{\beta} = (\mathbb{E}[\tilde{\mathbf{X}}_i \mathbf{X}_i'])^{-1} \mathbb{E}[\tilde{\mathbf{X}}_i Y_i]$$

How to Estimate the Parameters

- Collect $\mathbf{X}_i$ into an $n \times k$ matrix $\mathbb{X} = (\mathbf{X}'_1, \dots, \mathbf{X}'_n)$
- Collect $\mathbf{Z}_i$ into an $n \times \ell$ matrix $\mathbb{Z} = (\mathbf{Z}'_1, \dots, \mathbf{Z}'_n)$
- In-sample projection matrix produces fitted values:

$$\widehat{\mathbb{X}} = \mathbb{Z}(\mathbb{Z}'\mathbb{Z})^{-1}\mathbb{Z}'\mathbb{X}$$

- Fitted values of the regression of $\mathbb{X}$ on $\mathbb{Z}$.
- Matrix party trick: $\mathbb{X}'\mathbb{Z}/n = (1/n) \sum_i^n \mathbf{x}_i\mathbf{z}'_i \xrightarrow{P} \mathbb{E}[\mathbf{x}_i\mathbf{z}'_i]$.
- Take the population formula for the parameters:

$$\beta = (\mathbb{E}[\tilde{\mathbf{x}}_i\mathbf{x}'_i])^{-1}\mathbb{E}[\tilde{\mathbf{x}}_i y_i]$$

- And plug in the sample values (the n cancels out):

$$\widehat{\beta}_{2SLS} = (\widehat{\mathbb{X}}'\mathbb{X})^{-1}\widehat{\mathbb{X}}'\mathbf{y} \xrightarrow{P} \beta$$

- This is how R/Stata estimate the 2SLS parameters.

Asymptotic Variance for 2SLS

- We can write the centered, normalized TSLS estimator as:

$$\sqrt{n}(\widehat{\beta}_{2SLS} - \beta) = \underbrace{\left(n^{-1} \sum_i \widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'\right)^{-1}}_{\xrightarrow{p} (\mathbb{E}[\widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'])^{-1}} \underbrace{\left(n^{-1/2} \sum_i \widehat{\mathbf{x}}_i \varepsilon_i\right)}_{\xrightarrow{d} N(0, \mathbb{E}[\widehat{\mathbf{x}}_i' \varepsilon_i' \varepsilon_i \widehat{\mathbf{x}}_i])}$$

- Thus, $\sqrt{n}(\widehat{\beta}_{2SLS} - \beta)$ has asymptotic variance:

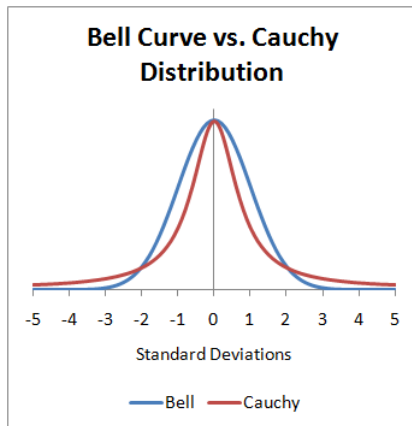
$$(\mathbb{E}[\widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'])^{-1} \mathbb{E}[\widehat{\mathbf{x}}_i' \varepsilon_i' \varepsilon_i \widehat{\mathbf{x}}_i] (\mathbb{E}[\widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i'])^{-1}$$

- **Robust 2SLS variance estimator** with residuals $\widehat{u}_i = Y_i - \mathbf{x}_i' \widehat{\beta}$:

$$\widehat{\text{var}}(\widehat{\beta}_{2SLS}) = (\widehat{\mathbb{X}}' \widehat{\mathbb{X}})^{-1} \left(\sum_i \widehat{u}_i^2 \widehat{\mathbf{x}}_i \widehat{\mathbf{x}}_i' \right) (\widehat{\mathbb{X}}' \widehat{\mathbb{X}})^{-1}$$

- HC2, clustering, and autocorrelation versions exist.

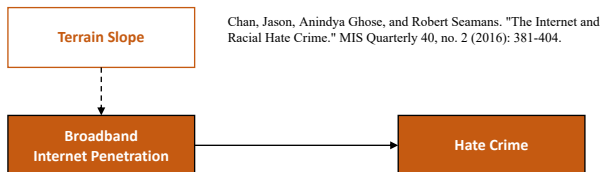
Cauchy vs. Normal Distribution



Source: <https://stats.stackexchange.com/questions/36027/why-does-the-cauchy-distribution-have-no-mean>

Regional Characteristics as IVs

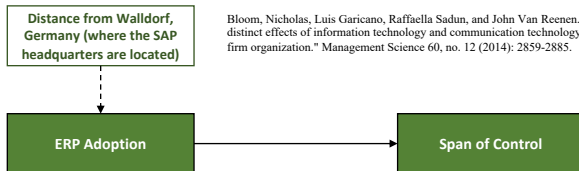
- Chan, Ghose, Seamans, MISQ, 2016:



- Observational study using *Hate Crime Statistics* from FBI.
 - RQ: Does the spread of Internet increase racial hate crime?
- Issue? Confounding
- IV? terrain slope/steepness (i.e., how many hills in a given region?)

Geographical Proximity-Based IVs

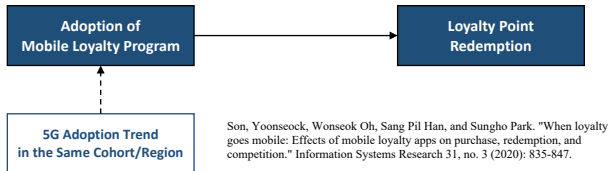
- RQ: Does the adoption of an ERP system impact plant manager autonomy (i.e., span of control)?
 - Span of control: the number of employees managed by supervisors or managers in an organization (high SoC = centralized).



- Observational study: the CEP management and organization survey and the Harte-Hanks ICT panel (Bloom, Garicano, Sadun, Van Reenen, *MgmtSci*, 2014).
- IV: Distance from the ERP market leader (i.e., SAP) w/ 25%+ market share $\rightsquigarrow$ likely more established connections with SAP (German firms vs. French, England firms).

Macro/Cohort Trends as IVs

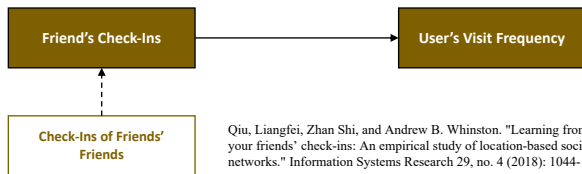
- RQ: How does loyalty app adoption affect customer point redemption behavior?



- Observational study: data on loyalty app adoption status, loyalty point redemption patterns, and purchase behaviors in multivendor loyalty program (MVLP) context (Son, Oh, Han, Park, ISR, 2020).
- IV: 5G adoption rate in the same cohort (e.g., age group).

Peers' Environments as IVs

- Qiu, Shi, Whinston, ISR, 2018:

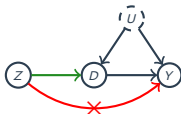


- Observational study using data on restaurant check-in information and the users' social network ties from a major SNS in China.
 - RQ: Is there observational learning/herding effect for restaurant discovery?
- Issue? Homophily.
- IV: Check-in activities of friend's friends.

Bartik (Shift-Share) IVs

Barron et al. (*Marketing Sci.* 2020)⁴

- Context: Panel of 43,000 ZIP codes (100 largest U.S. metro areas), monthly 2011-2016. Airbnb listings scraped; Zillow rent & price indices matched at ZIP-month.



Instrument Z: Google-Trends “Airbnb” (global shock) $\times$ 2010 ZIP “touristiness” (# food/lodging firms)

Treatment D: $\ln(1 + \text{Airbnb listings})$ in ZIP-month.

Outcome Y: Zillow rent index, house price index, price-to-rent.

- Identification issue: Hot ZIPs attract both Airbnb supply and rising housing costs $\rightsquigarrow$ upward bias.
- IV logic:
 - Conditional exogeneity: 2010 touristiness fixed before Airbnb’s national growth; Google-Trends shock is global, not driven by ZIP-specific factors. **(Hardest assumption for Bartik IVs; defended in detail next slide.)**
 - Relevance: first-stage $F = 650\text{--}820$ well above Stock-Yogo cutoff.
 - Exclusion: global search shocks unrelated to local housing; touristiness fixed in 2010. City $\times$ month FE + placebo ZIPs (no listings) show no direct price effect.
 - Monotonicity: more global Airbnb awareness cannot lower listings in touristy ZIPs.

⁴Barron, K., Kung, E., & Proserpio, D. “The Effect of Home-Sharing on House Prices and Rents” *Marketing Sci.* 2020. 40(2): 283–304.

Bartik (Shift-Share) IVs

Barron et al. (*Marketing Sci.* 2020)

- IV logic 1 (continued): defending **conditional exogeneity**, the hardest assumption for Bartik IVs:
 - Parallel pre-trends: prior to 2012, rents and prices evolve identically across touristiness quartiles (Fig. 5) $\rightsquigarrow$ no pre-existing divergence.
 - Touristiness $\times$ time test: adding a ZIP-specific trend term ($h_{i,2010} \times t$) yields an insignificant coefficient; IV effect unchanged.
 - Rich time-varying controls: results robust after adding ZIP income, population, hotel rooms, TripAdvisor reviews, and airport arrivals $\rightsquigarrow$ IV not picking up gentrification or tourism shocks.